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Neogene Tectonic Rotation of the Santa Ynez Range, Western Transverse Ranges, California, Suggested by Paleomagnetic Investigation of the Monterey Formation

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A paleomagnetic investigation was conducted on dolomite beds in the middle and upper Miocene Monterey Formation in order to determine the timing and amount of post-early Miocene tectonic rotation experienced by the Santa Ynez Range (part of the western Transverse Ranges province of California). Five stratigraphic sections of the Monterey Formation exposed along the coastline west of Santa Barbara were extensively sampled. The sections were correlated using tuff beds, and ages were assigned to each dolomite bed on the basis of magnetostratigraphy and diatom biostratigraphy. The dolomite beds possess stable primary remanent magnetic directions which pass fold tests and reversal tests, and show a consistent decrease in magnetic declination with decreasing age. Rocks 14.8 to 15.8 m.y. old have a mean magnetic declination of $92 \pm 7°$, whereas rocks 8 to 11 m.y. old have a mean magnetic declination of $36 \pm 7°$. The consistent upsection decrease in magnetic declination within these stratigraphic sections is attributed to a rapid $56 \pm 10°$ clockwise rotation of the entire Santa Ynez Range between 15.2 ± 0.1 and 10 ± 1 m.y. B.P. The mean magnetic declination of $36 \pm 7°$ in rocks 8 to 11 m.y. old is probably due to continued clockwise rotation of the Santa Ynez Range during the Plio-Pleistocene. An early Miocene palinspastic reconstruction is proposed in which the area of the presently east-west trending Santa Ynez Range is restored to an initially north-south orientation.

INTRODUCTION

The western Transverse Ranges comprise an east-west trending structural province which crosscuts the NW-SE structural grain of central and southern California. The province is bounded on the north by the Santa Ynez fault system, on the east by the San Gabriel fault, on the south by the Malibu Coast fault and the Northern Channel Islands, and on the west by the continental shelf (Figure 1). The Santa Ynez Range forms the northern margin of the province. The presently east-west structural trends in the western Transverse Ranges developed through a long series of deformational episodes which began in the Jurassic [*Jones et al.*, 1976], acted intermittently in the Paleogene and Miocene, and culminated in the Plio-Pleistocene [e.g., *Hoots*, 1931; *Dibblee*, 1950, 1966; *Weaver*, 1969]. As a consequence of this tectonic activity, the axis of Cenozoic deposition within the western Transverse Ranges is also presently east-west trending (the Santa Barbara Embayment [*Reed and Hollister*, 1936; *Edwards*, 1971; *Nagle and Parker*, 1971; *McCracken*, 1972; *Howell*, 1974]).

Hypotheses regarding the origin of the anomalous east-west structural orientation and stratigraphic trends with the western Transverse Ranges include (1) emplacement of the province by strike-slip faulting and counterclockwise rotation along a "paleo-Big Bend" in the North American–Pacific plate boundary [*Hall*, 1978, 1981*a*, *b*; *Dickinson*, 1979, 1982, 1983], (2) clockwise rotation of the terrain into its present orientation as a consequence of dextral shear between the North American and Pacific plates [*Hamilton*, 1978; *Kamerling and Luyendyk*, 1979; *Crouch*, 1979; *Luyendyk et al.*, 1980], (3) creation of the east-west trending features in situ by vertical tectonic processes [*Reed and Hollister*, 1936; *Dibblee*, 1950, 1966; *Baird et al.*, 1974].

Kamerling and Luyendyk [1979, this issue] and *Luyendyk et al.* [1980] have shown that the early and middle Miocene volcanics in the Santa Monica Mountains and the Northern Channel Islands have remanent magnetic directions which are deflected 14-138 degrees clockwise from the expected magnetic direction for the Miocene (Figure 1). They interpreted these clockwise-deflected paleomagnetic directions to result from clockwise rotation of the entire western Transverse Ranges province as a coherent tectonic block during the middle and late Miocene. Hall [1981*a*, *b*] has advocated an alternative interpretation that attributes the clockwise-deflected paleomagnetic declinations to local block rotations within the western Transverse Ranges.

These conflicting interpretations have dramatically different implications for the geologic history of the western Transverse Ranges. The hypothesis of *Dickinson* [1979, 1982] and *Hall* [1981*a*, *b*] implies that the large-magnitude right-lateral strike-slip displacements experienced by the Hosgri and Rinconada faults in the Coast Ranges were taken up by throughgoing right-lateral strike-slip displacement on the Santa Ynez fault during the Miocene. In contrast, the hypothesis of *Luyendyk et al.* [1980] implies that the right-lateral strike-slip on these Coast Range faults terminated against the Santa Ynez fault system during Miocene clockwise rotation of the Santa Ynez Range. During the Paleogene, the present area of the western Transverse Ranges was the site of a forearc basin [*Hamilton*, 1978; *Crouch*, 1979]. The interpretations of clockwise versus counterclockwise rotation imply opposite polarities for this forearc basin. If the western Transverse Ranges province was emplaced by counterclockwise rotation during Miocene right-lateral strike-slip displacement on the Santa Ynez fault, then Paleogene sediments in the Santa Ynez Range would have been deposited on the continentward side of the forearc basin. Alternatively, if the western Transverse Ranges were emplaced by clockwise rotation during the Neogene, then the Paleogene sediments in the Santa Ynez Range would

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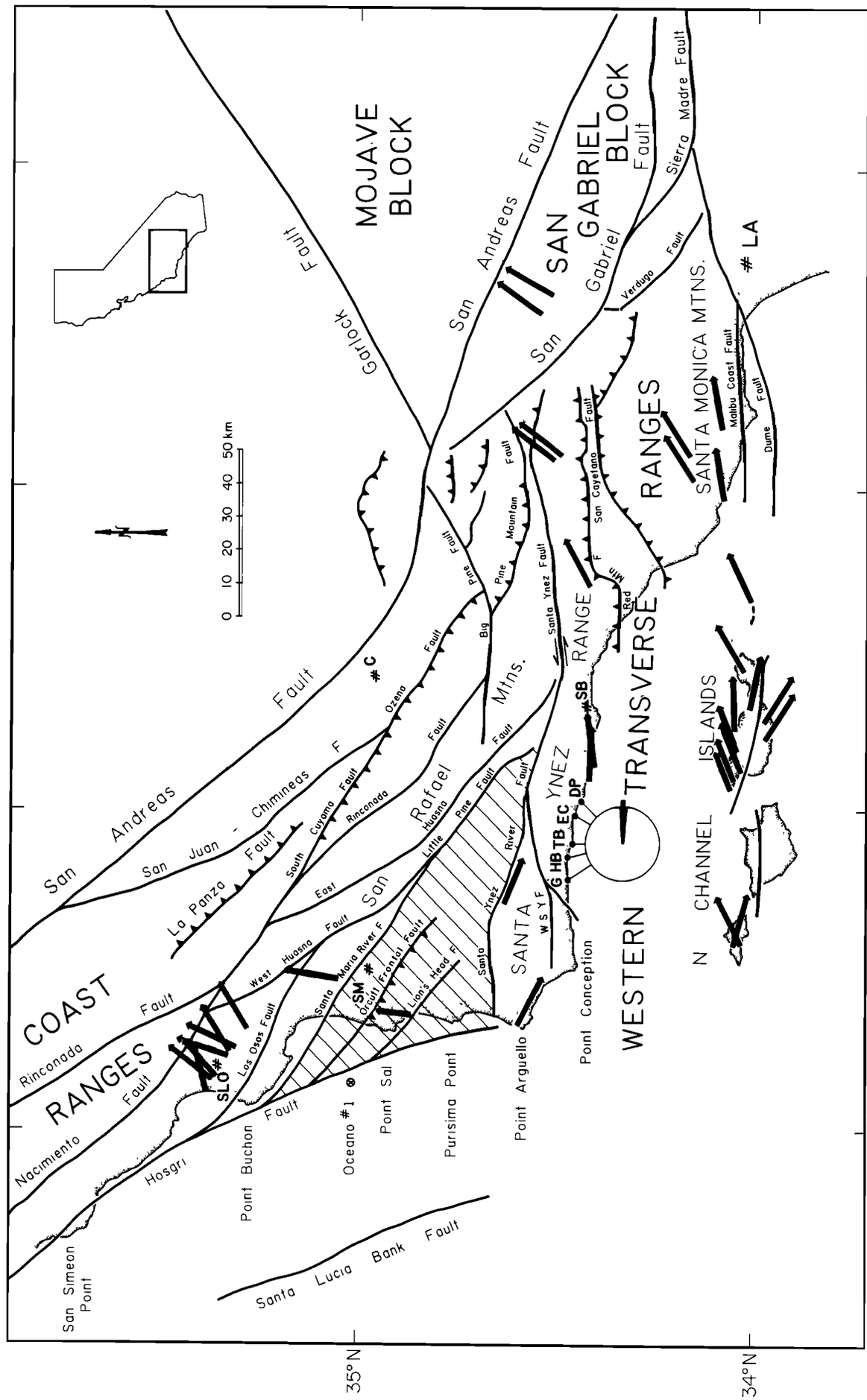


Fig. 1. Tectonic setting of the Santa Ynez Range within the western Transverse Ranges province of southern California. Eastward pointing arrow within the circle indicates the mean paleomagnetic declination observed in 14.8 to 15.8 m.y. old Monterey dolomites from the stratigraphic sections sampled in this study (DP = Dos Pueblos, EC = El Capitan, TB = Tajiguas Beach, HB = Hondo Beach, G = Gaviota). Other arrows indicate the paleomagnetic declinations obtained from early and middle Miocene rocks elsewhere in the Coast Ranges and western Transverse Ranges (sources: *Greenhaus and Cox* [1979], *Kamerling and Luyendyk* [1979], *Luyendyk et al.* [1980, Figure 1], and *Hornafius* [1984a]). Fault pattern from *Jennings* [1973] and *Dibblee* [1976]. Cross-hatched pattern = areal extent of the Santa Maria Basin. C = Cuyama, LA = Los Angeles, SB = Santa Barbara, SM = Santa Maria, SLO = San Luis Obispo, WSYF = western Santa Ynez fault.

have been deposited on the oceanward side of the forearc basin.

In order to discriminate between these alternative hypotheses, a paleomagnetic investigation was conducted on the middle and upper Miocene Monterey Formation exposed along the coastline between Santa Barbara and Point Conception. This study was intended to address the following questions:

1. Has the Santa Ynez Range experienced any tectonic rotation since the early Miocene? If so, is the rotation clockwise or counterclockwise in sense?
2. What is the maximum amount of post-early Miocene tectonic rotation in the Santa Ynez Range? Is this amount sufficient to align pre-Miocene structural and lithologic trends within the Santa Ynez Range with those in the Coast Ranges and Peninsular Ranges to the north and south?
3. How coherent are the tectonic rotations within the Santa Ynez Range? Did the range rotate as a single tectonic block, or did it rotate as smaller tectonic domains?
4. What is the timing of the rotation, and how rapidly did it occur?

GEOLOGIC SETTING

The Santa Ynez Range is an east-west trending antiform which was folded and uplifted during the Plio-Pleistocene [*Dibblee*, 1950]. The core of the antiform is cut by the Santa Ynez fault, which is a high-angle reverse fault with a component of left-slip. A continuous 7-km-thick upper Cretaceous through upper Miocene sedimentary sequence is exposed along the south-dipping homocline which comprises the southern flank of the Santa Ynez Range [*Dibblee*, 1966]. To the north of the Santa Ynez Range, within the Santa Maria basin, Paleogene strata are absent, and Miocene strata rest directly on Jurassic-Cretaceous Franciscan basement [*Woodring and Bramlette*, 1950]. The Santa Ynez River fault forms the boundary between the Paleogene-bearing strata in the Santa Ynez Range and strata lacking Paleogene sediments in the Santa Maria basin [*Sylvester and Darrow*, 1979].

The Monterey Formation in the Santa Ynez Mountains spans the middle and late Miocene in age ($\sim$16.5 m.y. to $\sim$6 m.y. old). The base of the Monterey Formation is separated from the underlying Rincon Shale by a tuff or bentonite bed, which *Dibblee* [1950] mapped as the lateral equivalent of the Tranquillon Volcanics. Two plagioclase separates from this tuff gave K-Ar ages of 16.1 $\pm$ 0.6 and 16.8 $\pm$ 0.5 m.y. [*Turner*, 1970]. The Monterey Formation is conformably overlain by the Sisquoc Formation, which is latest Miocene and early Pliocene in age [*Isaacs*, 1983].

Between Santa Barbara and Point Conception, five structurally intact stratigraphic sections of the Monterey Formation are well-exposed along the coastline (see Figure 1). The sections are in generally southward dipping homoclines (20°–70°), although one section (El Capitan) dips to the west (55°). The lower Monterey Formation is composed of calcareous and siliceous shales, whereas the upper Monterey Formation is mostly siliceous shale [*Dibblee*, 1950; *Isaacs*, 1980, 1983]. Within the lower Monterey Formation at Dos Pueblos Beach, J. Barron (personal communication, 1982) has documented the occurrence of a disconformity which ranges in age from 14 to 11 m.y. old. This disconformity occurs across a 10-m-thick phosphatic-pebble-conglomerate horizon in an organic-rich interval of the Monterey Formation (the Organic Shale Member of *Isaacs* [1980]). Phosphatic conglomerate horizons also occur in other stratigraphic sections of the Monterey Formation along the southern flank of the Santa Ynez Range, where they were also interpreted to represent disconformities (see below).

PROCEDURES AND METHODS

Paleomagnetic Procedures

Dolomite beds and calcite-cemented tuffs within the Monterey Formation possess a stable primary magnetic remanence [*Hornafius et al.*, 1981]. Other rock types are either remagnetized, unstably magnetized, or too weakly magnetized to be measured ($< 10^{-8}$ emu/cm^3). The dolomite occurs as both continuous beds and as nodules. The remanent magnetization in the dolomite is carried by pseudo-single-domain (PSD) detrital magnetite, and was probably acquired within a few meters of the seafloor [*Hornafius*, 1984*b*]. The highest-quality paleomagnetic results are obtained from well-indurated dolomites (density > 2.7 g/cm^3).

Cores were collected from the dolomite with a hand-held gas-powered diamond drill and were oriented with a Brunton compass and orienting device. The cores were drilled over as large an outcrop area as possible, and a separate tilt correction was measured for each core collected. Four to sixteen cores (diameter = 2.5 cm) were drilled from each dolomite bed, and the cores were then cut into 2.3-cm segments (= samples). The direction and intensity of magnetization of each sample was measured on the SuperConducting Technologies (SCT) cryogenic magnetometer at the University of California at Santa Barbara.

Each sample was either demagnetized in an alternating magnetic field with a Schonstedt GSD-1 single-axis demagnetizer, or thermally demagnetized in a box furnace with a nulled magnetic field (< 100 gammas). Alternating field (af) demagnetization was carried out in 50 or 100 oersteds (Oe) steps, until the magnetic intensity of the cores fell below 5×10^{-8} emu/cm^3. This usually occurred between 200 and 300 Oe. Thermal demagnetization was also conducted in a step-wise manner, at 100, 200, 250, and 300°C. Above 300°C, magnetic intensities usually increased, and directions became erratic, due to the creation of new magnetic phases [*Hornafius*, 1984*b*].

Samples were considered to be magnetically stable if they displayed less than a 5° change in direction per 50 Oe or 50°C, in the range 200–300 Oe or 200–300°C (Figure 2). Samples with stable magnetic directions within 20° of the dipole field direction for the sites ($I = 54°$, $D = 0°$, and $I = -54°$, $D = 180°$) were considered to be remagnetized (see below). Samples which had a total magnetic moment of less than 5×10^{-7} emu after 200 Oe or 200°C demagnetization were rejected, because their magnetic directions could not be reliably measured. Stable directions were usually attained by 200 Oe or 250°C (Table 1).

A separate tilt correction was applied to each stably magnetized sample, and the structurally corrected directions were then averaged to obtain the unit means reported in Table 1. Because an independently measured bedding correction was applied to each sample, an estimate of the tilt correction error is included in the alpha 95 for each bed. Unit means with an alpha 95 greater than 15° or with a precision parameter (kappa) of less than 50 were rejected.

A total of 440 cores were collected for this study: 299 cores (68%) possessed a stable primary magnetization, 72 cores (16%) possessed a stable secondary magnetization, and 69 cores (16%) were either unstably magnetized or very weakly

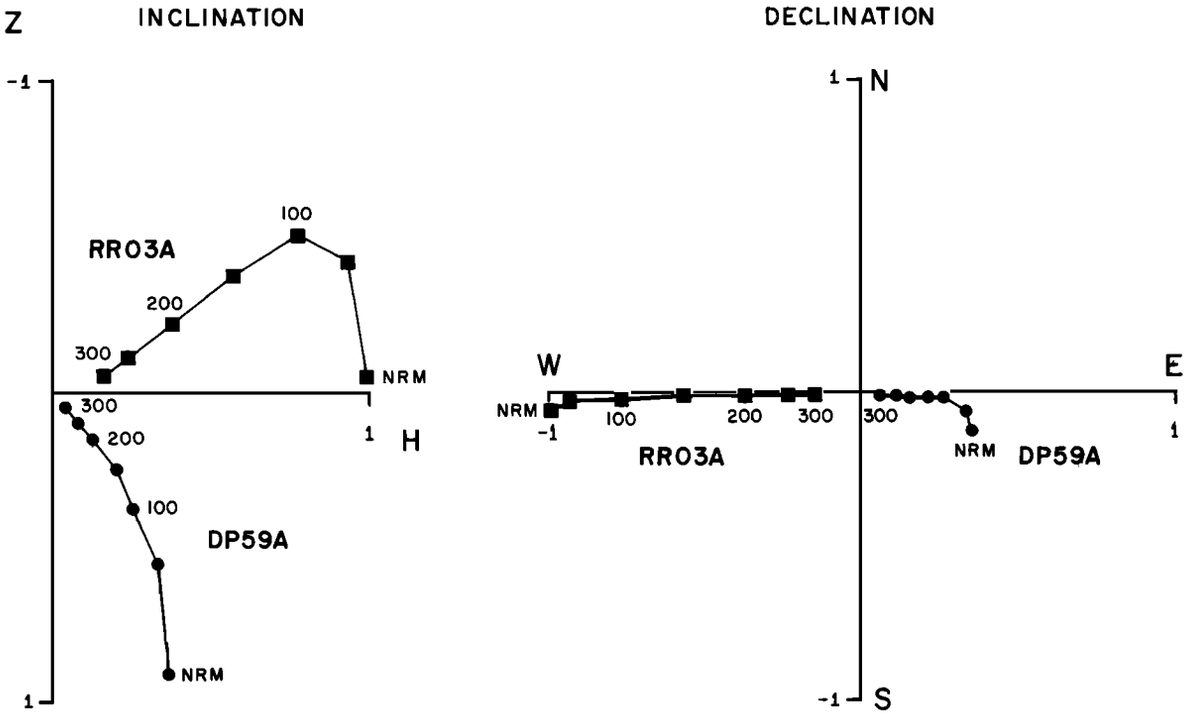


Fig. 2. Orthogonal projections of remanent magnetic vectors from two dolomite samples from the Monterey Formation, during step-wise af demagnetization (in Oe). Left: horizontal (H) versus vertical (Z) components of the structurally corrected magnetic vectors. Right: projection of the structurally corrected magnetic vectors onto the horizontal plane. RR03A is from a 15.4 m.y. old reversed-polarity bed in the Tajiguas Beach section. DP59A is from a 14.9 m.y. old normal-polarity bed in the Dos Pueblos Beach section. In both samples, a stable magnetic direction was attained by 200 Oe. The stable magnetic declinations in both samples are clockwise-deflected ~90° from the expected N-S direction.

magnetized. Out of the 72 remagnetized cores, 38 were remagnetized in the normal-polarity dipole field direction, and 34 were remagnetized in the reversed-polarity dipole field direction. The results of this work are based on 320 stably magnetized samples from 45 dolomite beds and 3 calcite-cemented tuff beds.

Stratigraphic Correlations

The measured stratigraphic sections used in this study are depicted in Figure 3. The sections were correlated with the Dos Pueblos Beach section using two distinctive sets of tuffs. The Dos Pueblos Beach section was used as a reference section because of the excellent biostratigraphic age control in that section. Correlation of the other stratigraphic sections with the Dos Pueblos Beach section enabled precise ages to be assigned to the dolomite beds sampled in these other stratigraphic sections (see below). The two sets of tuffs used for stratigraphic correlation occur above and below the 11–14 m.y. old disconformity in the Dos Pueblos Beach section.

The tuff set below the disconformity in the Dos Pueblos Beach section occurs 18 m above the base of the provincial Luisian stage as determined by *Kleinpell* [1938] and 6 m below the base of subzone b of the *Denticulopsis lauta* diatom zone (J. A. Barron, personal communication, 1982) (Figure 3), to which *Keller and Barron* [1981] have assigned an age of 14.7 m.y. This set of three to six tuffs is recognizable by the presence of ilmenite in the lower tuff beds in the set. In the Tajiguas Beach section (Figure 3), two tuff beds in the set are calcite cemented and yield stable reversely magnetized directions (Table 1). The magnetic stratigraphy in the Hondo Beach section (Figure 4) places these tuff beds at the top of the lowest reversed interval in paleomagnetic epoch 15 (14.74–14.90 m.y. B.P. [*Ness et al.*, 1980]). An age of 14.8 m.y. is assigned to these tuffs for the purposes of this paper.

Sixteen tuffs occur over a 40-m-thick stratigraphic interval above the disconformity in the Dos Pueblos Beach section, within the provincial Mohnian stage as determined by *Kleinpell* [1938]. These tuff beds were correlated with tuffs in the Tajiguas Beach and Gaviota Beach sections on the basis of their thickness and the stratigraphic spacing between the tuff beds. The poorly exposed upper Hondo Beach section was correlated with the Tajiguas Beach section on the basis of lithologic comparisons alone. This particular correlation is therefore considered to be the least reliable of the correlations which were made.

Age Assignments

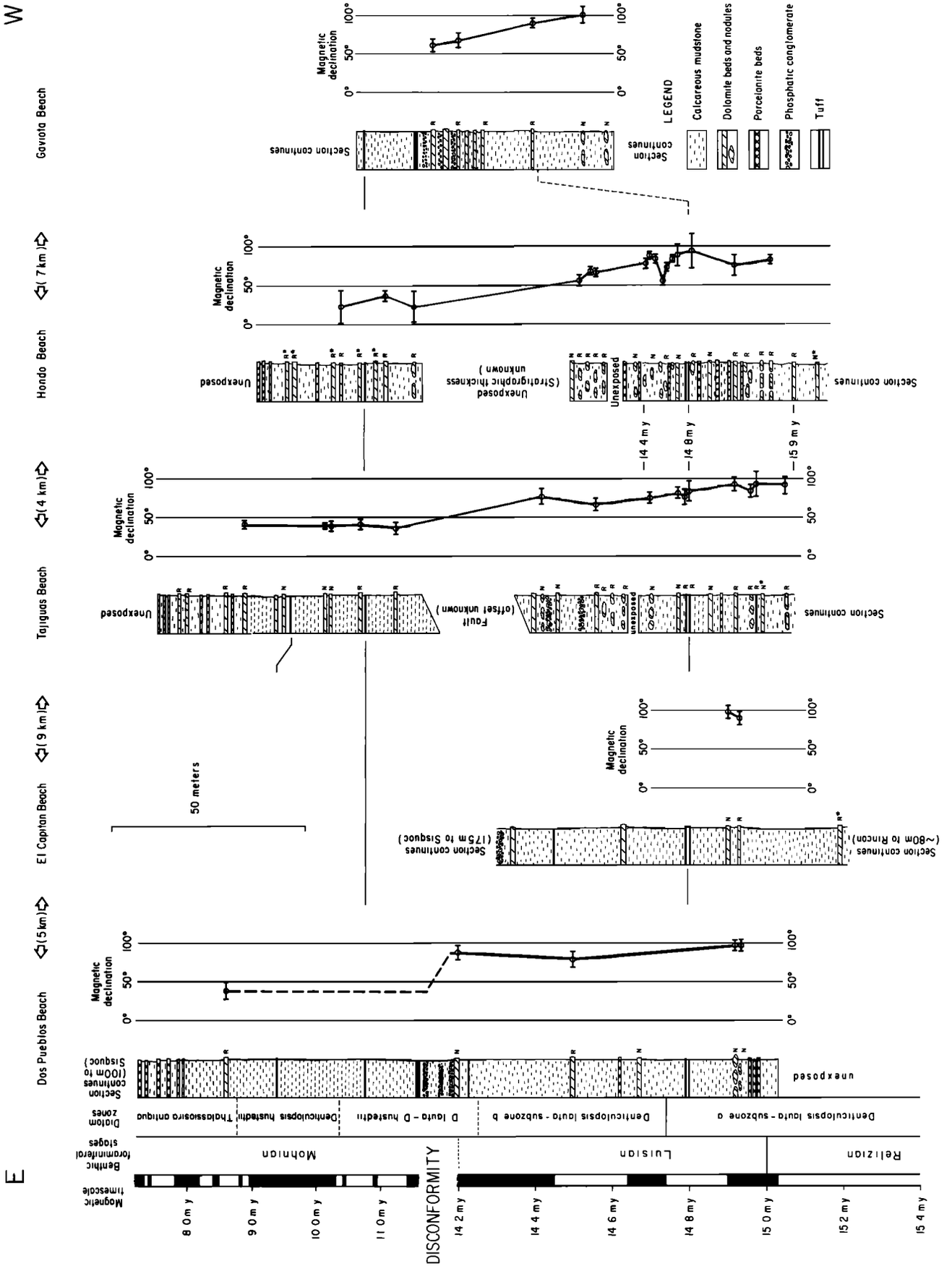
Each dolomite bed used in this study was assigned an age based on diatom biostratigraphy and/or magnetostratigraphy (Table 1). The age assignments for the beds below the disconformity in the Dos Pueblos Beach section are based on the range of subzone b of the *Denticulopsis lauta* diatom zone and the magnetic polarity of the beds (Figure 3). Correlation of the El Capitan Beach section with the Dos Pueblos Beach section suggests that the R to N polarity transition recorded by the dolomite beds sampled at El Capitan Beach corresponds with the magnetic epoch 15/16 boundary (15.03 m.y. B.P.; Figure 3). The prominent dolomite beds in the lower parts at the Tajiguas and Hondo Beach sections can be matched up one-for-one. The age assignments for both of these stratigraphic sections is based on the magnetic polarity sequence in the Hondo Beach section (Figure 4). *Annellus californicus* v. *hannai*

TABLE 1. Paleomagnetic Directions From Dolomite Beds in the Monterey Formation (Corrected to Normal Polarities)

Stratigraphic Height, m	Number of Samples	Mean Inclination, deg	Mean Declination, deg	α_{95}	κ	Age, m.y.	Polarity	Demagnetization Level, Oe or °C
			Dos Pueblos Beach: Latitude = 34.4°N, Longitude = 120.0°W					
10	9	45.2	96.5	5.2	100.1	14.9	N	200 Oe/250°C
11	11	49.1	97.1	4.2	117.4	14.9	N	200 Oe/250°C
53	5	43.4	77.9	7.3	109.6	14.5	R	250°C
86	7	47.8	87.2	5.9	104.8	14.2	N	200 Oe/250°C
Mean*	4	46.7	89.5	7.7	141.8	14.9–14.2		
130	6	42.9	38.6	5.3	158.8	8.5	R	250°C
			El Capitan Beach: Latitude = 34.5°N, Longitude = 120.0°W					
26	7	47.8	90.3	5.4	126.0	15.0	R	200 Oe/200°C
29	6	25.1	97.6	7.4	83.5	15.0	N	200 Oe
Mean*	2	36.5	94.5	53.4	24.0	14.9–15.1		
			Tajiguas Beach: Latitude = 34.5°N, Longitude = 120.1°W					
0	5	45.4	92.1	7.9	94.9	15.8	R	200 Oe
8.4†	4	51.0	93.6	9.9	87.3	15.5	R	250°C
9.2	5	34.9	84.1	6.6	134.1	15.4	R	200 Oe/250°C
13.0	7	45.3	92.9	6.4	91.3	15.3	R	200 Oe
Mean*	4	44.2	90.2	8.5	118.6	15.8–15.3		
26.0†	7	53.5	82.6	7.9	59.9	14.8	R	300°C
26.3†	6	24.8	77.2	9.4	51.9	14.8	R	300 Oe/250°C
27.5	15	49.7	81.3	4.6	70.1	14.7	N	200 Oe/250°C
35.0	6	51.1	75.0	5.0	177.2	14.4	N	200 Oe/250°C
Mean*	4	44.9	78.8	15.6	35.7	14.8–14.4		
8‡	6	42.8	67.2	5.7	140.4	11–14	R	200 Oe
22‡	9	48.5	76.9	6.9	57.1	11–14	N	200 Oe/250°C
Mean*	2	45.8	71.8	19.4	167.9	11–14		
6‡	9	46.6	36.2	5.4	91.9	~11.0	R	250°C
14‡	8	44.6	40.9	4.2	176.9	~10.5	R	250°C
23‡	11	47.6	38.8	3.8	144.6	~10.0	N	250°C
24‡	6	44.2	39.7	3.1	457.3	~10.0	N	250°C
44‡	5	60.1	41.9	4.0	376.0	~8.5	R	300°C
Mean*	5	48.6	39.4	6.4	145.8	8.5–11.0		
			Hondo Beach: Latitude = 34.5°N, Longitude = 120.1°W					
12.5	9	50.6	83.3	3.8	180.6	15.7	R	200 Oe/250°C
22.8	5	46.6	75.6	9.4	66.9	15.2	R	200 Oe/250°C
33.2	3	49.5	95.1	14.9	69.4	14.8	R	200 Oe
Mean*	3	49.2	84.5	10.4	142.0	15.7–14.8		
37.0	5	50.7	89.6	9.1	72.1	14.7	N	200 Oe/250°C
37.4	6	54.4	85.1	2.4	753.9	14.7	N	200 Oe/250°C
39.8	10	47.4	73.3	3.6	181.6	14.6	R	200 Oe/250°C
40.7	16	47.2	55.8	3.7	102.5	14.5	R	200 Oe/250°C
42.5	6	51.9	84.8	3.2	447.5	14.5	N	200 Oe/250°C
44.2	4	50.8	89.1	2.7	1165.3	14.4	N	200 Oe/200°C
44.9	9	54.4	78.9	3.8	181.4	14.4	N	200 Oe/200°C
Mean*	7	51.5	79.2	6.1	98.0	14.7–14.4		
58.2	5	49.4	67.6	2.9	711.4	11–14	R	200 Oe
59.2	5	50.3	68.7	3.0	669.0	11–14	R	200 Oe/200°C
62.1	4	56.2	56.9	3.8	586.2	11–14	R	200 Oe/250°C
Mean*	3	52.1	64.8	8.2	227.5	11–14		
0‡	3	45.4	23.8	14.6	72.3	~11.5	R	200 Oe
7‡	6	40.4	37.3	7.3	84.7	~11.0	R	200 Oe/250°C
18‡	4	56.1	25.3	13.0	51.1	~10.5	R	250°C
Mean*	3	47.5	29.3	14.6	72.2	10.5–11.5		
			Gaviota Beach: Latitude = 34.5°N, Longitude = 120.2°W					
6	11	44.1	100.5	6.5	50.2	~14.9	N	200 Oe/250°C
19	7	46.0	79.0	4.3	196.0	~14.7	R	200 Oe
37	6	41.8	67.3	6.6	104.7	11–14	R	200 Oe
44	6	52.0	61.4	4.9	188.0	11–14	R	200 Oe
Mean*	2	46.9	64.6	24.1	109.4	11–14		

*Average of mean directions from individual beds (for the specified time interval).
†Tuff beds from the Tajiguas Beach stratigraphic section.
‡Measured from the top of the unexposed intervals at Tajiguas and Hondo Beach (Figure 3).

was recovered from the lowest reversed-polarity dolomite bed in the Gaviota Beach section (C. W. Lagle, personal communication, 1983) (Figure 3). Since this diatom is restricted in occurrence to the top of magnetic epoch 16 and the bottom of magnetic epoch 15 [*Opdyke et al.*, 1974; *Barron*, 1981], the normal and reversed polarity dolomite beds at the base of the Gaviota Beach section (Figure 3) were assigned to the lowest normal and reversed polarity events in paleomagnetic epoch


W
E
Gaviota Beach
Hondo Beach
Tajiguas Beach
El Capitan Beach
Dos Pueblos Beach
(7 km)
(4 km)
(9 km)
(5 km)
Magnetic declination
100°
50°
0°
LEGEND
Calcareous mudstone
Dolomite beds and nodules
Porcelanite beds
Phosphatic conglomerate
Tuff
Section continues
Unexposed
Unexposed (Stratigraphic thickness unknown)
Fault (offset unknown)
14 4 m y
14 8 m y
15 9 m y
50 meters
Section continues (175m to Sisquoc)
Section continues (~80m to Rincon)
Section continues (100m to Sisquoc)
Diatom zones
Thalassiosira antiqua
Denticulopsis hustedtii
D. lauta - D. hustedtii
Denticulopsis lauta - subzone b
Denticulopsis lauta - subzone a
Benthic foraminiferal stages
Mohnian
Luisian
Relizian
Magnetic timescale
DISCONFORMITY
8 0 m y
9 0 m y
10 0 m y
11 0 m y
14 2 m y
14 4 m y
14 6 m y
14 8 m y
15 0 m y
15 2 m y
15 4 m y

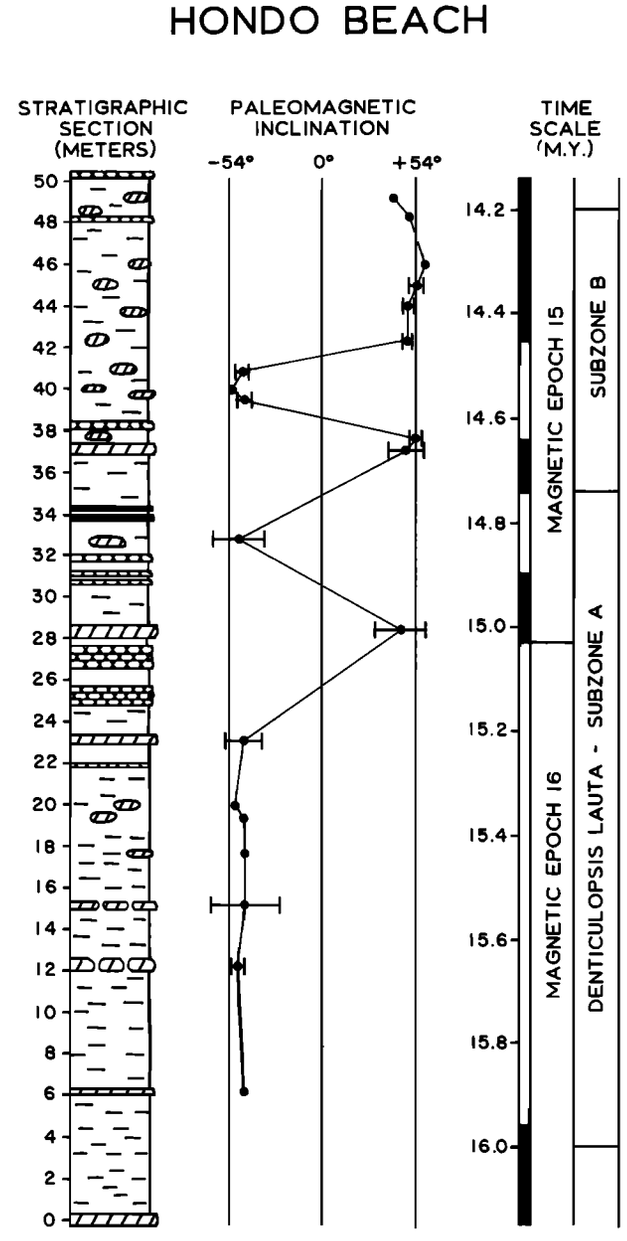


Fig. 4. Magnetic inclination vs. stratigraphic height in the lower Hondo Beach section ($\Delta I = \alpha_{95}$). Correlation of the polarity sequence in this stratigraphic section with the magnetic timescale [from *Ness et al.*, 1980] enabled the age of the 14 to 16 m.y. old dolomite beds the Hondo and Tajiguas Beach sections to be precisely determined (± 0.1 m.y.; Table 1). The tuff beds at 34 m in the stratigraphic section were correlated with the tuff beds in the Dos Pueblos, El Capitan, and Tajiguas Beach sections (Figure 3). This figure includes magnetic directions from beds with α_{95} greater than 15°, as well as directions from individual samples, and therefore includes more results than are listed in Table 1 and shown in Figure 3. The bed at the base of the measured section is remagnetized in the present-day field direction, so the normal-polarity primary direction predicted for this bed could not be confirmed. Correlation of the *Denticulopsis lauta* diatom zone with the magnetic timescale is from *Keller and Barron* [1981]. Diatom zonation of this section indicates that the base of subzone b of the *D. lauta* zone is at 37 m in the section [*Hornafius*, 1984*a*], in agreement with the correlations of *Keller and Barron* [1981]. The postcompaction sedimentation rate during lower paleomagnetic epoch 15 in this section is ~25 m/m.y., compared with a postcompaction sedimentation rate of ~100 m/m.y. in the Dos Pueblos Beach section during the same time interval (Figure 3).

Fig. 3. Magnetic declination versus stratigraphic height in the Monterey Formation exposed along the coastline west of Santa Barbara. Each stratigraphic section displays a consistent upsection decrease in magnetic declination between rocks ~15 and ~11 m.y. old (Table 1). The stratigraphic sections were correlated with two distinctive sets of tuff beds (see text). The ages assigned to the Dos Pueblos Beach section are based on diatom zonation of the section by J. Barron (personal communication, 1982), and the magnetic polarity of the dolomite beds (using the magnetic time scale of *Ness et al.* [1980]). Benthic foraminiferal stage assignments are from Kleinpell (1938). N = normal polarity bed. R = reversed polarity bed. Asterisk = remagnetized bed (normal or reversed). Declinations of the reversed-polarity beds were corrected to normal-polarity declinations. Error bars were calculated from $\Delta D = \sin^{-1}[\sin \alpha_{95}/\cos I]$.

15 (Figure 4). The accuracy of the age assignments for the 14 to 16 m.y. old dolomite beds used in this study is considered to be ±0.1 m.y. (Table 1).

The age assignments for the stratigraphic interval above the disconformity in the Dos Pueblos Beach section are based on the range of the *Denticulopsis hustedtii* diatom zone, to which *Barron* [1981] has assigned an age of 8.4–10.2 m.y. The age assignments for the dolomite beds in the upper Tajiguas Beach and upper Hondo Beach sections are based on stratigraphic correlation with the Dos Pueblos Beach section (Figure 3). The accuracy of the age assignments for the 8 to 11 m.y. old dolomite beds is considered to be ±1.0 m.y. (Table 1). Strata in the Tajiguas, Hondo, and Gaviota Beach sections occur between rocks that are 11 and 14 m.y. old, and are therefore equivalent in age to the 11 to 14 m.y. old disconformity in the Dos Pueblos Beach section. The age of the dolomite beds within these 14 to 11 m.y. old stratigraphic intervals could not be more precisely determined (Table 1).

PALEOMAGNETIC RESULTS

Magnetic Declinations

All primary paleomagnetic directions obtained from the Monterey are significantly clockwise-deflected from the expected Miocene direction after structural correction (Table 1 and Figure 5). Rocks 14.8 to 15.8 m.y. old have a mean magnetic declination of 92 ± 7°, and the mean magnetic declination decreases with decreasing age (Table 2). This systematic decrease in magnetic declination with decreasing age is attributed to a rapid clockwise rotation of the Santa Ynez Range during the middle Miocene.

In Figure 3, the mean (structurally corrected) paleomagnetic declination from each bed is plotted against stratigraphic height. At all sites, the declinations show a dramatic decrease in strata between 11 and 14 m.y. old. The paleomagnetic declination also appears to decrease upsection in strata ranging in age from 14 to 16 m.y. old, which suggests that tectonic rotation was proceeding rapidly during deposition of these stratigraphic intervals. The lack of an upsection decrease in declination in beds 8–11 m.y. old in the Tajiguas and Hondo Beach sections suggests that the rotation slowed considerably by the beginning of the late Miocene.

Table 2 gives temporal means of the paleomagnetic directions obtained from the study area, averaged over different time intervals. A mean age is assigned to each regional mean direction, based on the ages assigned to each bed in Table 1. Beds 14.8 to 15.8 m.y. old have a regional mean declination of 92 ± 7° (mean age = 15.2 m.y.), whereas beds 8–11 m.y. old have a mean declination of 36 ± 7 (mean age ~10 ± 1 m.y.). If the earth's magnetic field was an axial dipole, these results imply that the study area rotated 55.8 ± 10.0° clockwise during the middle Miocene (15–10 m.y. ago), at an average rate of ~10°/m.y.

The time of initiation of rotation and the total amount of rotation experienced by the Santa Ynez Range is unconstrained by the paleomagnetic data presented in this paper. The regional paleomagnetic mean directions from beds 14.8–15.8 m.y. old and beds 14.2–14.8 m.y. old are significantly different (Table 2 and Figure 6), which suggests that rotation was proceeding rapidly by 15 m.y. ago. The initiation of rotation probably coincided with the initiation of wrench faulting in the Salinian block and the southern California borderland approximately 16 m.y. ago [*Graham*, 1978; *Crouch*, 1981] (see below). If so, the total amount of clockwise rotation experienced by the Santa Ynez Range during the Neogene would not exceed 100° (Figure 6).

The mean declination of 36° observed in rocks 8–11 m.y. old suggests that the rotation event was not completed by the end of the middle Miocene. If rotation continued into the late Miocene at a rate of 10°/m.y., the rotation would have been completed by the end of the late Miocene. However, the paleomagnetic evidence from the upper Monterey Formation in the Santa Ynez Range [*Hornafius*, 1984*a*] suggests that the remaining ~30° of rotation did not occur until the Plio-Pleistocene (Figure 6).

Magnetic Inclinations

Primary paleomagnetic inclinations from the Monterey dolomites are consistently shallower than the expected dipole field inclination of 54° for the study area at its present latitude (Table 1). Compared to the Miocene reference pole for western North America [*Irving*, 1979], the magnetic inclination of the 15.2 and 10 m.y. regional means are too shallow by 10.0 ± 7.0° and 7.1 ± 7.0°, respectively (Table 2). Assuming that the earth's magnetic field was a centered axial dipole during the Miocene, these inclination anomalies imply a northward translation of 8.1° and 4.8° since 15.2 and 10 m.y. B.P., respectively. Post-early Miocene displacement along the San Andreas fault (=315 km in a northwest direction [*Matthews*, 1976]) would account for 2° of northward translation. If the entire Santa Ynez Range rotated 90° clockwise about an eastern pivot (with respect to North America), this would account for an additional 1° northward translation since 15 m.y. B.P. These two effects are therefore insufficient to account for the amounts of northward translation suggested by the paleomagnetic inclinations, using the centered dipole field hypothesis. Geological relationships in southern California make an undetected northward translation of 3 to 5 degrees extremely unlikely [see *Crowell*, 1973], since this would require that 200 to 400 km more slip has occurred on the San Andreas fault system during the Neogene than has been suggested. The shallow magnetic inclinations observed in the Monterey Formation may be due instead to an off-centered dipole [*Wilson and McElhinney*, 1974; *Harrison and Watkins*, 1979] or to nondipole field behavior [*Cox*, 1975; *Coupland and Van der Voo*, 1980] during the Miocene (see discussion by *Luyendyk et al.* [this issue]). Since the off-centered dipole (or a dipole with zonal nondipolar components) is still axially symmetric [*Wilson*, 1971; *McElhinny*, 1973], this departure from the centered dipole field assumption would not significantly affect the interpretation of the declination data.

Remagnetized Directions

At each site the primary paleomagnetic directions from beds of approximately the same age pass a reversal test. Since each site has a different tectonic tilt correction, the primary magnetic directions from beds of the same age also pass regional fold tests (Figure 5). In contrast, beds with structurally uncorrected directions within 20° of the present-day dipole field direction failed the fold test (Figure 7). These secondary directions are therefore inferred to have been acquired subsequent to middle Pleistocene folding of the Monterey Formation in the Santa Ynez Range [*Dibblee*, 1950, 1966].

Remagnetized directions from any one stratigraphic interval were either entirely normal or entirely reversed in polarity. The reversed secondary directions must have been acquired prior to the Brunhes normal polarity epoch (>0.72 m.y. ago [*Ness et al.*, 1980]). Since the regional mean secondary direc-

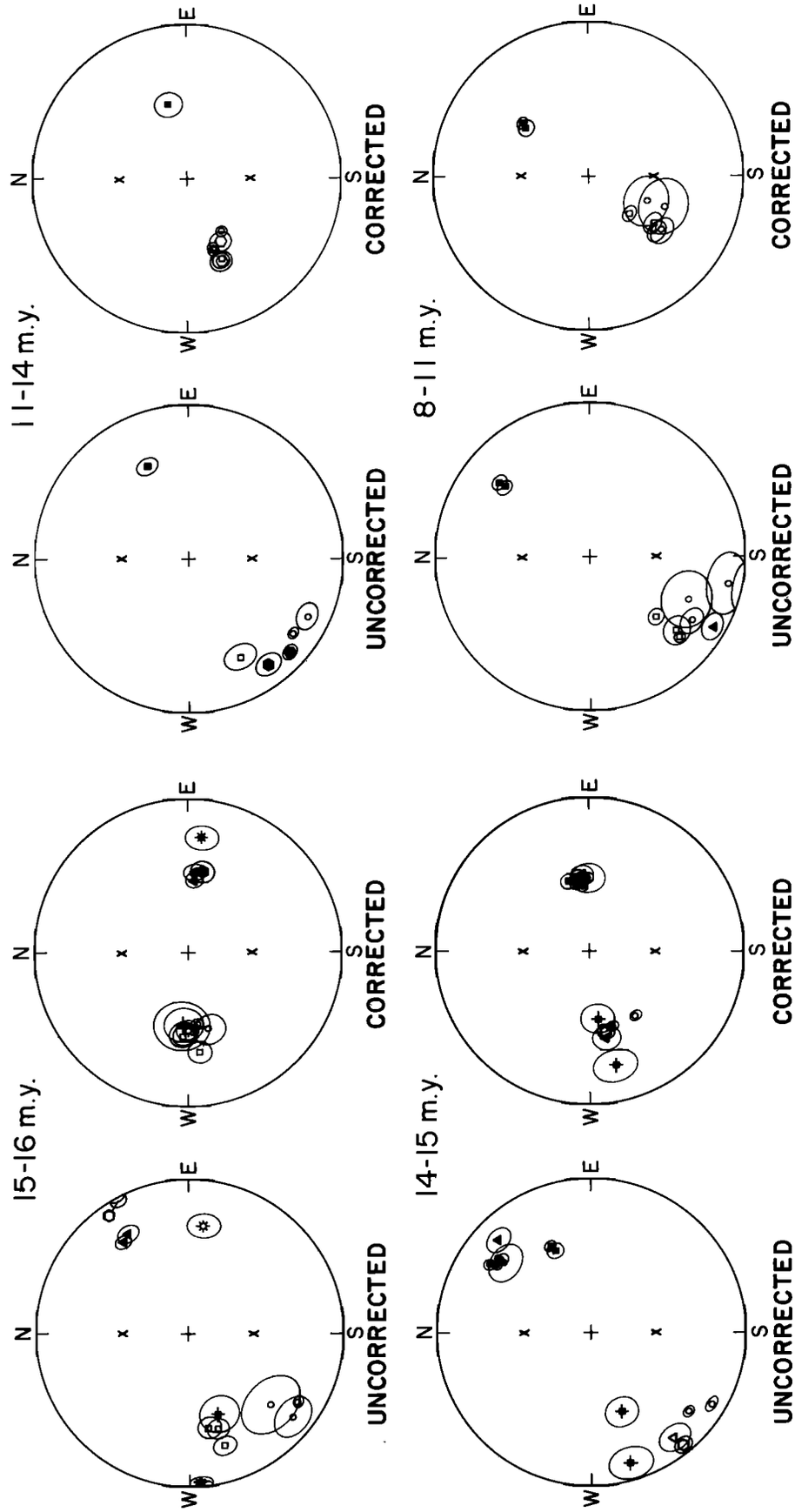


Fig. 5. Equal-area stereographic projections of the structurally corrected and uncorrected mean magnetic directions listed in Table 1, plotted according to age. These stable primary magnetic directions pass fold tests and reversal tests and show a consistent decrease in magnetic declination with decreasing age (Table 2). The error ellipses indicate the uncertainty in the mean directions at the 95% confidence level. Open symbols = upper hemisphere. Solid symbols = lower hemisphere. Triangles = Dos Pueblos Beach. Stars = El Capitan Beach. Squares = Tajiguas Beach. Circles = Hondo Beach. Hexagons = Gaviota Beach. Crosses = calcite-cemented tuff beds from Tajiguas Beach. X = present-day dipole field direction. The 15–16 m.y. plot includes all beds below the 14.8 m.y. old tuff beds (Figure 3).

TABLE 2. Mean Paleomagnetic Directions From Monterey Formation Averaged According to Age

Age Range, m.y.	Mean Age,* m.y.	Number of Beds Averaged	Mean Inclination, deg	Mean Declination, deg	α_{95}	κ	VGP Latitude, deg	VGP Longitude, deg	Rotation† R	Flattening† F
14.8–15.8	15.2	12	44.8	91.7	4.9	80.7	13.3	−53.2	95.2 ± 9.2	10.0 ± 7.0
14.2–14.8	14.6	14	48.4	79.6	4.6	75.6	24.1	−50.3	83.1 ± 9.3	6.4 ± 6.8
14.2–15.8	14.9	26	46.9	85.3	3.6	64.4	19.0	−51.7	88.8 ± 8.1	7.9 ± 6.2
11–14	?	7	48.8	66.9	4.8	159.8	34.2	−45.2	70.4 ± 9.5	6.0 ± 6.9
8–11	~10	9	47.7	35.9	4.9	109.7	59.0	−29.5	39.4 ± 9.5	7.1 ± 7.0
5–8‡	~6.5	(4)§	52.4	31.3	9.5	94.0	63.8	−36.2	34.8 ± 16.8	2.4 ± 10.7
0.7–1.2¶	?	6	49.8	1.2	9.1	55.3	86.0	45.4	5.0 ± 16.5	3.5 ± 11.6

*Mean age calculated from the age assigned to each bed in Table 1.

†R and F calculated from the equations of *Beck* [1980], relative to the stable North American pole of *Irving* [1979] except for the 0.7–1.2 mean, which was calculated relative to the worldwide Plio-Pleistocene and Quaternary (0–5 m.y.) pole of *McElhinny* [1973].

‡From *Hornafius* [1984*a*].

§The mean is the average of four sites from the upper Monterey Formation in the Santa Ynez Range; four to eight beds were averaged to obtain each site mean.

¶Remagnetized directions from Table 3.

tion of the reversely magnetized beds is not significantly different from the expected direction (Table 3), the southern flank of the Santa Ynez Range could not have experienced a significant amount of tectonic rotation or tilt (>15°) during the late Pleistocene to Recent.

DISCUSSION

The consistently clockwise-deflected magnetic declinations observed in the lower Monterey Formation suggest that 95 ± 9° of clockwise rotation has occurred along the southern flank of the Santa Ynez Range since 15.2 m.y. ago, relative to stable North America (Table 2). During the middle Miocene (15.2–10 m.y. ago), the southern flank of the range rotated through 56 ± 10°. The rotation appears to have slowed or stopped in the late Miocene, and then resumed at a rapid rate in the Plio-Pleistocene (Figure 6).

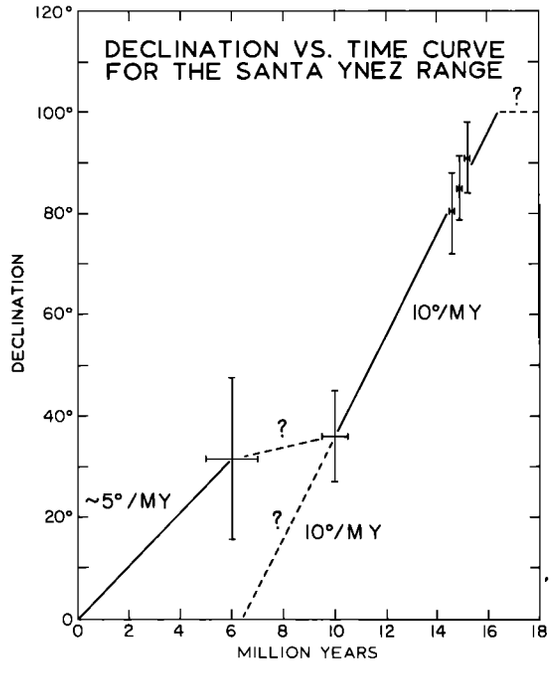


Fig. 6. Declination versus age curve for the Santa Ynez Range (from Table 2). The data suggest that the Santa Ynez Range experienced rapid clockwise rotation during the middle Miocene and Plio-Pleistocene, but that little rotation occurred during the late Miocene. The time of initiation of rotation (~16 m.y. B.P.) is based on geological evidence. The inflection points correspond in age with major tectonic changes which occurred throughout western North America at 16, 10, and 5 m.y. B.P. [see *Barrash and Venkatakrishnan*, 1982]. Declination error bars calculated from $\Delta D = \sin^{-1}[\sin \alpha_{95}/\cos I]$.

The southern flank of the Santa Ynez Range is not cross-cut by a systematic network of strike-slip faults, which would be required in order for coherent local block rotations to occur [*Freund*, 1974]. The decrease in magnetic declination with decreasing age observed in the Monterey Formation therefore implies that the rotation is regional in extent, and is not due to local block rotations, as suggested by *Hall* [1981*a*, *b*]. Since the study area is not separated from the rest of the Santa Ynez Range by major strike-slip faults, the entire range probably participated in the rotation suggested by the paleomagnetic data from its southern flank. This conclusion is corroborated by paleomagnetic data from the remainder of the Santa Ynez Range [*Hornafius*, 1984*a*].

The middle Miocene rotation occurred during deposition of the Monterey Formation in the areas now occupied by the Santa Ynez Range, the Santa Maria Basin, and the Santa Barbara Channel (Figure 1). Along the southern flank of the Santa Ynez Range, the middle Miocene rotation corresponds in age to a disconformity within the Monterey Formation. The disconformity is recognizable as a phosphatic-pebble-conglomerate horizon between strata 11 and 14 m.y. old, extending from Santa Barbara to Point Conception. No angular

TABLE 3. Reversed-Polarity Secondary Paleomagnetic Directions From the Monterey Formation (Structurally Uncorrected), After 300-Oe af Demagnetization

Location	Stratigraphic Height, m	Number of Samples	Mean Inclination	Mean Declination	α_{95}	κ
El Capitan	0	6	−48.8	195.2	4.9	186.9
Hondo	9	4	−53.9	182.6	4.0	514.9
Hondo	13	6	−45.6	191.9	6.0	126.2
Hondo	19	4	−53.6	182.1	7.8	139.3
Hondo	20	4	−36.2	169.3	5.8	253.4
Hondo	32	6	−57.6	165.0	5.8	132.0
Mean		6	−49.8	181.2	9.1	55.3

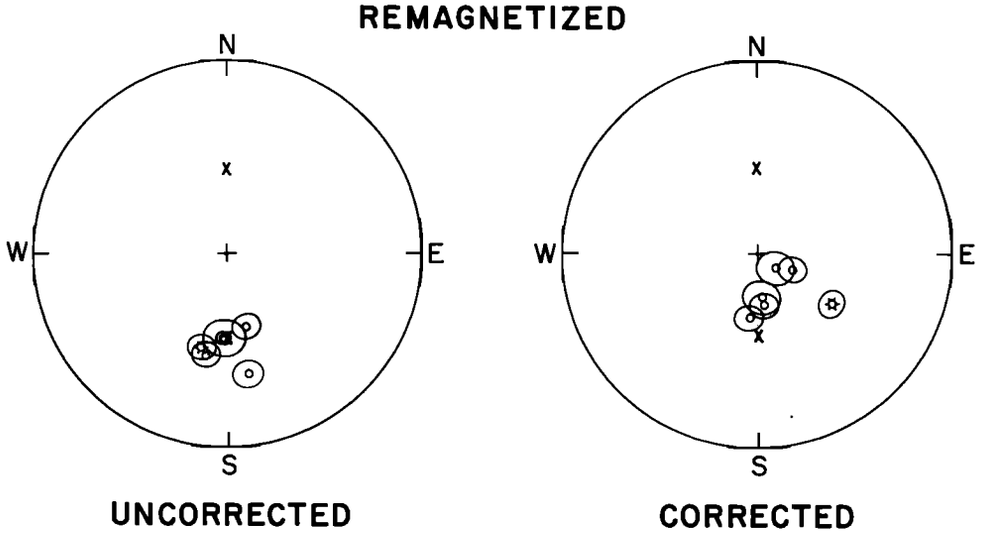


Fig. 7. Fold test on remagnetized beds. (Left) Equal-area stereographic projection of all mean magnetic directions which were within 20° of the reversed-polarity dipole field direction before structural correction (from Table 3). (Right) Magnetic directions from the same beds after structural correction. Since the dispersion in the magnetic directions increases after structural correction, these beds are inferred to be remagnetized. Symbols same as in Figure 5.

discordance is observed across this disconformity. The middle Miocene rotation was therefore not accompanied by significant tectonic compression within the Santa Ynez Range.

Palinspastic Restoration of the Santa Ynez Range

Neogene rotation of the Santa Ynez Range could have occurred in place, or during northward translation of the range as part of a western Transverse Ranges microplate. *Crouch* [1979] and *Kamerling and Luyendyk* [1979] have suggested that the Santa Ynez Range rotated clockwise during northward translation of the range from an initial position off Baja California. However, the idea of suturing of the Santa Ynez Range to the Coast Ranges along the Santa Ynez fault during the Neogene is inconsistent with the great similarities in stratigraphy between the Santa Ynez Range and the San Rafael Mountains [see *Dibblee*, 1966]. In particular, (1) the similarity of the regionally unique Eocene Juncal Formation [*Page et al.*, 1951] and Matilija Sandstone [*Link*, 1975] on both sides of the Santa Ynez fault, and (2) the correspondence between clasts in the Oligocene Sespe Formation and their inferred source areas to the north of the Santa Ynez fault [*McCracken*, 1972] argue for a close spatial relationship of the eastern Santa Ynez Range and the southern San Rafael Mountains since the Paleogene. It is therefore concluded that the Santa Ynez Range has rotated in place, with respect to the Coast Ranges, as suggested by *Luyendyk et al.* [1980].

Approximately 185 km of post-early Miocene strike-slip displacement has been postulated along the NW-trending faults in the Coast Ranges, to the west of the San Andreas fault (Table 4). Right-slip displacement has been suggested on the San Gregorio-Hosgri fault (115 km [*Graham and Dickinson*, 1978]), Rinconada fault (56 km [*Dibblee*, 1976]), and San Juan-Chimineas fault (14 km [*Dibblee*, 1976; *Bartow*, 1978]). All these faults terminate by splaying against the northern boundary of the western Transverse Ranges (Figure 1). *Freund* [1974] demonstrated that a strike-slip fault can terminate by splaying against a boundary, if the boundary rotates during active strike-slip displacement on the fault. *Luyendyk et al.* [1980] employed this mechanism in proposing a geometric model for rotation of the western Transverse Ranges, in which right-lateral strike-slip displacement on the NW-trending faults in the Coast Ranges and Peninsular Ranges is accommodated by clockwise rotation at the latitude of the Transverse Ranges. The boundaries of the rotated terrain were inferred to be the east-west trending faults bounding the Transverse Ranges province. This prediction is confirmed by the paleomagnetic data now available from the Transverse Ranges and the southern California borderland [*Luyendyk et al.*, this issue]. In the case of the Santa Ynez Range, the northern boundary of tectonic rotation is the Santa Ynez River fault [*Hornafius* 1984*a*] (see Figure 1).

In the early Miocene palispastic reconstruction in Figure 8, 60 km of right-lateral displacement has been removed from the San Gabriel fault [*Crowell*, 1975; *Ehlig et al.*, 1975]. The eastern Santa Ynez Range (between Santa Barbara and the San Gabriel fault) was then backrotated 60° [*Hornafius*, 1984*a*], and the western Santa Ynez Range (between Santa Barbara and Point Arguello) was backrotated 90°. Right-lateral strike-slip displacements on the NW-trending faults in the Coast Ranges were then removed. The 14 km of left-slip experienced by the Big Pine fault was restored in order to realign the Rinconada-Little Pine and Ozena-San Guillermo offset fault pairs to initially linear traces [*Vedder and Brown*,

TABLE 4. Suggested Post–Early Miocene Strike-Slip Displacements on Coast Range Faults

Fault	Displacement, km	Reference
San Gregorio–Hosgri fault	115	*Graham and Dickinson* [1978]
Sur-Nacimiento fault	0	*Graham* [1978]
Rinconada fault	56	*Dibblee* [1976]
San Juan–Chimineas fault	14	*Dibblee* [1976] and *Bartow* [1978]
Total suggested displacement	185	
Predicted displacement	210	equation (1)
Continuum shear (?)	25	(see text)

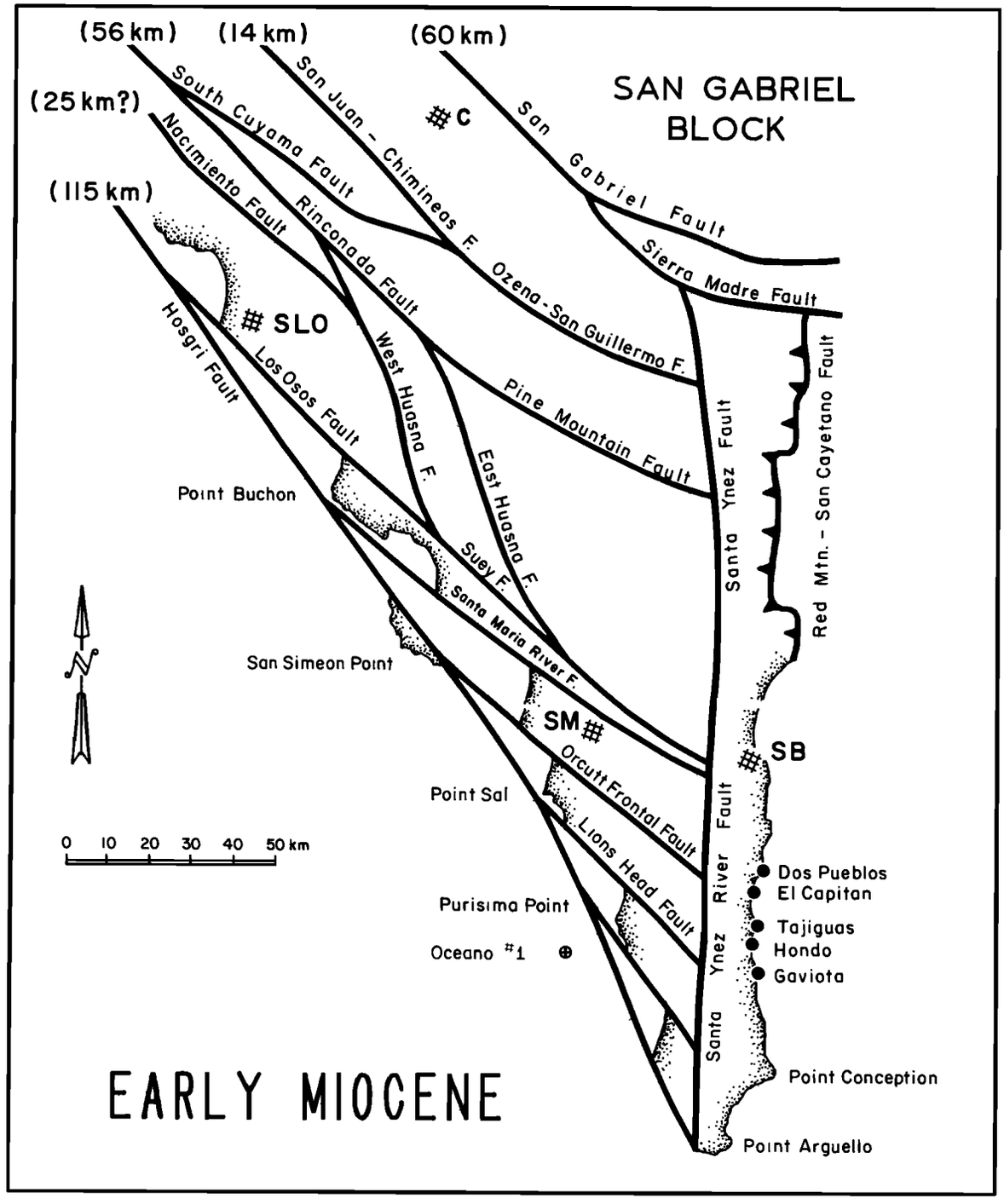


Fig. 8. Proposed early-Miocene palinspastic reconstruction of the Santa Ynez Range, relative to the southern California Coast Ranges and the San Gabriel block. Numbers in parentheses indicate the amount of suggested strike-slip displacement experienced by Coast Range faults since the early Miocene (Table 4).

1968]. The displacement at the southern end of the Hosgri fault was partitioned between the Los Osos fault, Santa Maria River fault, Orcutt Frontal fault, and Lion's Head fault, following *Hall* [1981*a*, *b*]. Point San Simeon and the Oceano 1 well were back-slipped 100 km along the Hosgri fault in order to bring them into juxtaposition with Point Sal and Purisima Point, respectively, as suggested by *Hall* [1981*a*, *b*]. Left-lateral offset on the Santa Ynez fault system since the early Miocene was assumed to be small, following *Dibblee* [1982]. The 75° clockwise rotation suggested for the Santa Monica Mountains and Northern Channel Islands implies that they occupied a position to the east of the Santa Ynez Range during the early Miocene [*Kamerling and Luyendyk*, this issue] (see Figure 9*a*).

Although paleomagnetic data from rocks older than 16 m.y. B.P. are not yet available to directly assess the total amount of clockwise rotation experienced by the Santa Ynez Range, the timing of initiation of clockwise rotation in the Santa Ynez

Range should correspond with the timing of initiation of wrench faulting to the north and south of the Transverse Ranges. Geological evidence from the southern California Coast Ranges and the San Joaquin Valley suggests that Neogene wrench faulting did not begin until ~16 m.y. B.P. (at the Saucesian-Relizian boundary), and that no wrench faulting occurred in these areas between the Eocene and the middle Miocene [*Clarke and Nilsen*, 1973; *Harding*, 1976; *Graham*, 1978]. The amount of clockwise rotation suggested by the paleomagnetic results from the Monterey Formation is therefore considered to be close to the total amount of clockwise rotation experienced by the Santa Ynez Range since the Eocene. If this is the case, then the early Miocene palinspastic reconstruction presented in Figure 8 should apply to the early Tertiary as well (Figure 9*a*).

Early Tertiary Paleogeography of the Santa Ynez Range

During the early Tertiary, a 5-km-thick arkosic sandstone and shale sequence was deposited along the southern flank of the Santa Ynez Range [*Dibblee*, 1966]. These sediments were deposited in a forearc-basin setting by submarine fan-delta complexes [*van de Kamp et al.*, 1974; *Hamilton*, 1978; *Link and Welton*, 1982] during oblique subduction of the Farallon plate beneath the North American plate [*Atwater and Molnar*, 1973]. The sedimentary sequence along the southern flank of the Santa Ynez Range records a history of nearly continuous deposition throughout the Paleogene, whereas the comparatively thin Paleogene sedimentary sequence along the northwestern margin of the range is interrupted by three unconformities [see *Dibblee*, 1950]. This change in continuity occurs across the western portion of the Santa Ynez fault (Figure 1), which appears to have acted as a hinge-line for the Eocene basin [*O'Brien*, 1973]. The complete absence of Eocene strata in the Santa Maria basin (see Figure 1) suggests that the basin was a subaerially exposed topographic high during the Eocene (the San Rafael High of *Reed and Hollister* [1936]). The Eocene shoreline is inferred to have followed the present trace of the Santa Ynez River fault along the southern margin of the Santa Maria Basin, and the Nacimiento-Huasna-Little Pine fault trend to the northeast of the basin [*Howell*, 1974].

A 90° clockwise rotation of the Santa Ynez Range since the Eocene implies that the early Tertiary forearc trough initially formed with its axis trending in a N-S direction. The sense of rotation further implies that the present area of the northwestern Santa Ynez Range formed the western (oceanward) margin of the early Tertiary forearc basin. The crustal slivers presently comprising the Santa Maria basin would have occupied a position to the west of the Santa Ynez Range prior to Neogene tectonic rotation of the western Transverse Ranges (Figure 8). The accretionary melange represented by the Franciscan basement in the Santa Maria basin was therefore in the correct position to form the trench-slope break for the early Tertiary forearc trough now preserved in the Santa Ynez Range. Repeated deformation along the northwestern margin of the Santa Ynez Range during the Paleogene is consistent with the inferred proximity of this area to the tectonically unstable trench-slope break.

Removing the effects of Neogene clockwise rotation thus restores the early Tertiary depositional trends within the Santa Ynez Range to an initial configuration which is consistent with the geometry of depositional systems along present-day convergent plate margins [see, e.g., *Dickinson and Seely*, 1979]. However, in order for the pre-Miocene (presently) east-west structural and depositional trends within the Santa Ynez Range to have formed with an initial northwest-southeast orientation, a 130° clockwise rotation would be required. Clockwise rotation of this magnitude does not appear to have occurred in the Santa Ynez Range, at least since 16 m.y. B.P. Initially N-S trending faults may therefore have cross-cut the forearc basin in the western Transverse Ranges during the early Tertiary.

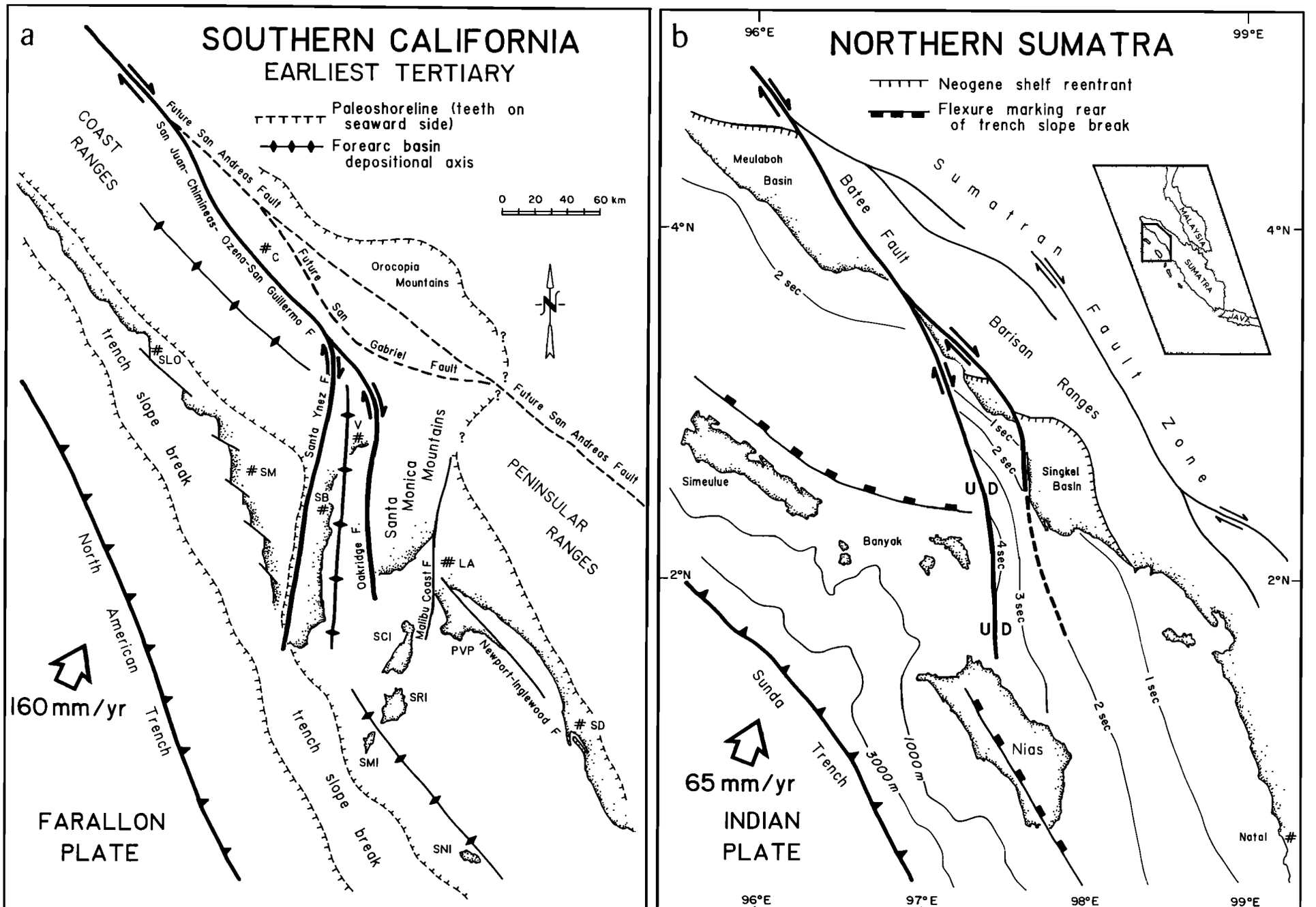
Faults that trend obliquely across the trend of a forearc basin are found today in Sumatra near the island of Nias (Figure 9). These cross-cutting faults (the Batee fault system of *Karig et al.* [1980]) are splays of the Sumatran fault zone, a right-lateral intra-arc fault system driven by oblique convergence between the Indian and Eurasian plates [see *Karig et al.*, 1979]. In California, oblique convergence between the Farallon and North American plates during the latest Cretaceous and earliest Tertiary resulted in 100–200 km of right-lateral displacement along the proto-San Andreas fault in northern California [*Graham*, 1978; *Dickinson*, 1983]. However, the locus of proto-San Andreas movement in southern California has not been identified [*Crowell*, 1973]. By analogy with the Sumatra example, it is suggested that some of the displacement on the proto-San Andreas fault in northern California was accommodated by right-lateral strike-slip on the initially N-S trending faults bounding the Santa Ynez Range; in particular, the Santa Ynez fault system and the Oakridge fault trend (Figure 9). In this interpretation, the ancestral Santa Maria basin occupied a position analogous to the Banyak Islands along the Sumatran trench-slope break, and the Santa Monica Mountains formed a shelf reentrant analogous to the Singkel Basin (Figure 9).

Extension Versus Compression During Rotation

Neogene tectonic rotation of a rigid Santa Ynez Range block would have resulted in crustal extension or compression along the margins of the rotated block. *Luyendyk et al.* [1980] suggested that basins opened at the joins between the rotated and unrotated blocks during rotation of the entire western Transverse Ranges province. According to their model, the middle Miocene basins adjacent to the Transverse Ranges resulted from the formation of deltoid-shaped gaps at the intersection of the NW trending right slip faults and the presently E-W trending faults bounding the western Transverse Ranges. In contrast, continued tectonic rotation within the western Transverse Ranges during the Plio-Pleistocene was accompanied by tectonic compression within the province. In order to understand this change from extension accompanying middle Miocene rotation to compression accompanying Plio-Pleistocene rotation, it is instructive to consider the geometry controlling deformation adjacent to a single strike-slip fault.

Whether extension or compression occurs along a transform boundary which consists of a single strike-slip fault is determined by the orientation of the fault trace relative to the slip vector for the fault. In the case of a right-lateral strike-slip fault, if the fault trace steps to the right (Figure 10*a*) then a releasing geometry exists which causes crustal extension and the formation of a pull-apart basin [*Crowell*, 1974]. On the other hand, if the fault trace steps to the left, then a restraining geometry exists, and crustal compression results (Figure 10*c*). In both cases, the resulting change in crustal area (ΔA) is simply related to the size of the step in the fault trace (s) and the amount of displacement experienced by the fault (d), $\Delta A = sd$.

By analogy with the foregoing principle, it is suggested that


a
SOUTHERN CALIFORNIA
EARLIEST TERTIARY
Paleoshoreline (teeth on seaward side)
Forearc basin depositional axis
0 20 40 60 km
COAST RANGES
Future San Andreas Fault
San Juan-Chimineas-Ozena-San Guillermo F
Future San Gabriel Fault
Orocopia Mountains
Future San Andreas Fault
PENINSULAR RANGES
trench slope break
Santa Ynez F
Oakridge F.
Santa Monica Mountains
Malibu Coast F
Newport-Inglewood F
North American Trench
160 mm/yr
FARALLON PLATE
trench slope break
C
SLO
SM
SB
V
LA
SCI
PVP
SD
SRI
SMI
SNI
b
NORTHERN SUMATRA
96°E
99°E
Neogene shelf reentrant
Flexure marking rear of trench slope break
Meulaboh Basin
Batee Fault
Sumatran Fault Zone
Barisan Ranges
MALAYSIA
SUMATRA
JAVA
4°N
2°N
2 sec
1 sec
2 sec
Simeulue
U D
Singkel Basin
Banyak
4 sec
3 sec
U D
Sunda Trench
3000 m
1000 m
65 mm/yr
INDIAN PLATE
Nias
1 sec
2 sec
Natal
96°E
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99°E

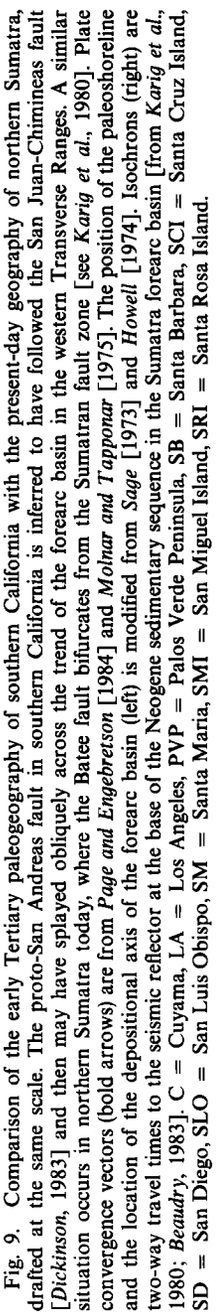


Fig. 9. Comparison of the early Tertiary paleogeography of southern California with the present-day geography of northern Sumatra, drafted at the same scale. The proto-San Andreas fault in southern California is inferred to have followed the San Juan-Chimineas fault [*Dickinson*, 1983] and then may have splayed obliquely across the trend of the forearc basin in the western Transverse Ranges. A similar situation occurs in northern Sumatra today, where the Batee fault bifurcates from the Sumatran fault zone [see *Karig et al.*, 1980]. Plate convergence vectors (bold arrows) are from *Page and Engebretson* [1984] and *Molnar and Tapponar* [1975]. The position of the paleoshoreline and the location of the depositional axis of the forearc basin (left) is modified from *Sage* [1973] and *Howell* [1974]. Isochrons (right) are two-way travel times to the seismic reflector at the base of the Neogene sedimentary sequence in the Sumatra forearc basin [from *Karig et al.*, 1980; *Beaudry*, 1983]. C = Cuyama, LA = Los Angeles, PVP = Palos Verde Peninsula, SB = Santa Barbara, SCI = Santa Cruz Island, SD = San Diego, SLO = San Luis Obispo, SM = Santa Maria, SMI = San Miguel Island, SRI = Santa Rosa Island.

the manner in which distributed shear was transferred across the western Transverse Ranges (i.e., from the southern California borderland to the Coast Ranges) controlled whether extension or compression accompanied tectonic rotation of the western Transverse Ranges. For example, if each NW-trending fault in the southern California borderland transferred its displacement to a fault to the northeast in the Coast Ranges, then gapping would result where each fault terminated against the boundary of the rotating block (Figure 10*b*). On the other hand, if each fault transferred its displacement to the left across the rotating block, compression would result between the offset fault pairs (Figure 10*d*). The change in crustal area resulting from each offset fault pair is again given by $\Delta A = sd$, as in the case of a through-going strike-slip fault.

The change from extension during the middle Miocene rotation to compression during the Plio-Pleistocene rotation suggests that the fault geometry in southern California changed from a releasing geometry to a restraining geometry at the end of the Miocene. This change in fault geometry was caused by the eastward jump in the position of the Pacific-North American transform boundary from the southern California borderland to the Gulf of California at the end of the Miocene [*Atwater and Molnar*, 1973]. During the middle Miocene, the strike-slip displacement experienced by the northern San Andreas fault [*Huffman*, 1972] was transferred across the rotating western Transverse Ranges province to the southern California borderland. The releasing geometry of this fault pattern caused the formation of the middle Miocene basins in southern California. At the end of the Miocene, strike-slip displacement began to occur on the southern San Andreas fault [*Ehlig et al.*, 1975] due to initiation of seafloor spreading in the Gulf of California [*Moore and Curray*, 1982]. Since some of the displacement experienced by the southern San Andreas fault was transferred to the Hosgri fault [*Bird and Rosenstock*, 1984], a restraining geometry was established, and compressional deformation accompanied tectonic rotation of the western Transverse Ranges during the Plio-Pleistocene.

Neogene Tectonic History of the Santa Maria Basin

At the scale of individual basins, whether extension or compression occurs during rotation of the western Transverse Ranges depends upon the location of the "contact points" between the rotated and unrotated blocks. A contact point is a point along the boundary of the rotating block at which no compression or extension occurs perpendicular to the boundary of the rotating block. For the northern margin of the rotating Santa Ynez Range block, rotation about contact points at the southwestern ends of the Coast Range blocks would produce gaps between the rotated and unrotated blocks (Figures 10*b* and 11), whereas rotation about contact points at the southeastern ends of the Coast Range blocks would produce overlaps (Figures 10*d* and 12). If the rotation occurred about contact points at the centers of the Coast Range blocks, the amount of gapping and overlapping would be equal, and no net extension or compression would result. Splaying of the NW trending faults where they terminate against the rotating block would also reduce the amount of gapping or overlapping [*Freund*, 1974].

During the Miocene, the Santa Maria basin formed where the Hosgri fault splays against the northern margin of the Santa Ynez Range [*Hall*, 1981*a*, *b*]. The resulting basin consisted of NW trending crustal slivers separated by right-lateral strike-slip faults, comparable to the geometry depicted in

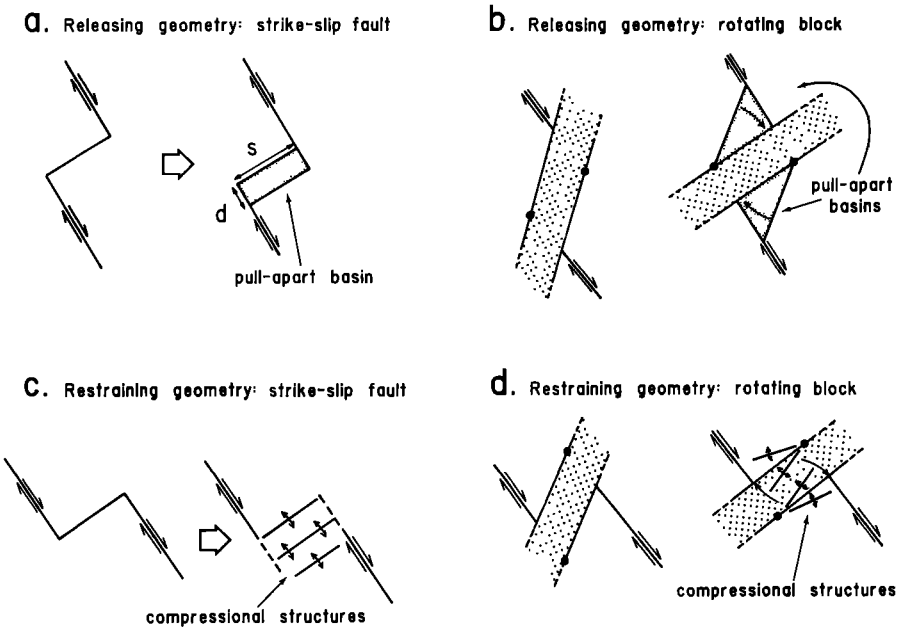


Fig. 10. Schematic diagram illustrating the manner in which extension or compression occurs during tectonic rotation. (*a*) Right-lateral strike-slip on a fault which steps to the right by a distance (*s*) and experiences a displacement (*d*) produces an increase in crustal area ($\Delta A = sd$), resulting in a pull-apart basin. (*b*) Similarly, right-lateral strike-slip on faults which step to the right across a cross-cutting block (stippled) produces deltoid-shaped gaps at the terminations of the faults and rotation of the intervening block. The position of the contact points for the rotating block relative to the adjacent nonrotating blocks are indicated by the dots. (*c*) Right-lateral strike-slip on a fault which steps to the left a distance (*s*) and experiences a displacement (*d*) produces a decrease in crustal area ($\Delta A = sd$), resulting in compressional deformation. (*d*) Similarly, right-lateral strike-slip on faults which step to the left across a cross-cutting block produces rotation and compressional deformation within the intervening block. The change from tectonic extension accompanying middle Miocene rotation of the western Transverse Ranges to tectonic compression accompanying the Plio-Pleistocene rotation is due to an eastward jump in the position of the Pacific-North American transform boundary from the southern California borderland (case *b*) to the Gulf of California (case *d*) at the end of the Miocene.

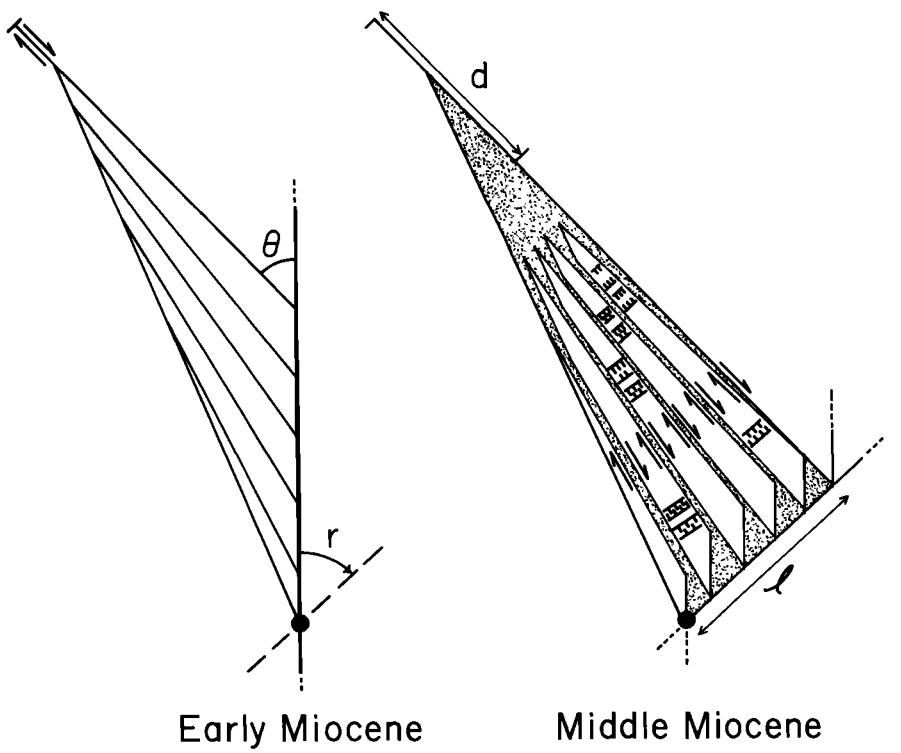


Fig. 11. Idealized fault geometry at the southern terminus of the Hosgri fault before and after middle Miocene clockwise tectonic rotation of the Santa Ynez Range (~16 and ~11 m.y. B.P.). The initial fault geometry is from Figure 8. Termination of middle Miocene strike-slip displacement on the Hosgri fault occurred by splaying of the fault against the northern boundary of the rapidly rotating Santa Ynez Range. Tectonic rotation about a contact point (=dot) at the western end of the Santa Ynez Range caused middle Miocene extension within the Santa Maria basin.

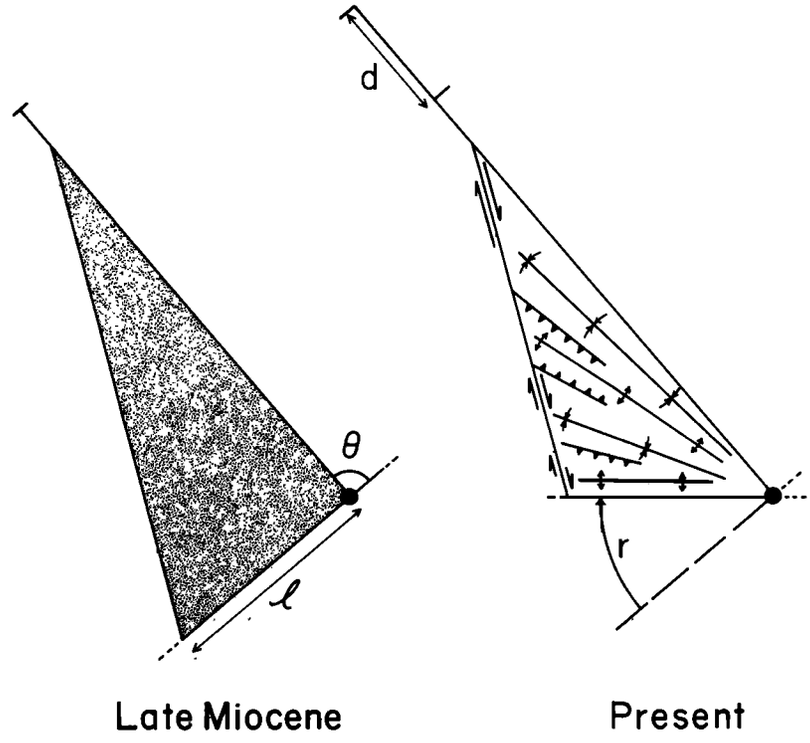


Fig. 12. Development of the present-day fault geometry and fold pattern within the Santa Maria basin, as a consequence of post-Miocene clockwise rotation of the Santa Ynez Range about an eastern contact point (dot). Termination of Plio-Pleistocene strike-slip displacement on the Hosgri fault was accomplished by N-S compressional deformation within the Santa Maria basin, to the east of the fault.

Figure 11. Crustal extension also resulted in elongation of these crustal slivers along NE trending normal faults within the basin [*California Division of Oil and Gas*, 1974]. This episode of crustal extension was followed by tectonic shortening and uplift within the basin during the Plio-Pleistocene [*Woodring and Bramlette*, 1950]. The Plio-Pleistocene rotation produced a radiating fold pattern within the Santa Maria basin, in which the fold axes eminate from the Plio-Pleistocene contact point at the southeast corner of the basin, in a manner analogous to that depicted in Figure 12. Termination of Plio-Pleistocene displacement on the Hosgri fault was accomplished by N-S directed shortening within the onshore Santa Maria basin, to the east of Hosgri fault, and the absence of N-S shortening to the west of the fault [*Payne et al.*, 1978].

Timing and Amount of Displacement on the Hosgri Fault

According to the *Freund* [1974] and *Luyendyk et al.* [1980] hypothesis, the amount of right-slip experienced by the NW trending faults in the Coast Ranges is dependent upon the amount of rotation experienced by the Santa Ynez Range. *Luyendyk et al.* [1980] presented equations which relate the amount of strike-slip on a particular fault to the amount of rotation and the width of the blocks bounding the fault. However, because of the splaying and anastomozing of faults at the southern end of the Coast Ranges (Figure 1), the width of the crustal blocks in the Coast Ranges is not a quantifiable parameter. It is preferable to relate the amount of strike-slip on a fault to the length of the rotating block (l) against which the fault, or set of faults, terminates (Figures 11 and 12). If r is the amount of rotation and θ is the initial angle between the NW-trending fault and the boundary of the rotated block, the amount of displacement on the NW-trending fault is given by

$$d = l[\cos\theta - \cos(r + \theta)] \quad (1)$$

This equation simply describes the amount of displacement between opposite ends of a lever arm (the rotated block) in a direction parallel to the slip vector for the fault (assumed to be oriented N40°W).

Splays of the Hosgri fault terminate against the entire length of the Santa Ynez River fault ($l = 100$ km). A 90° rotation of Santa Ynez Range since the early Miocene ($\theta = 40°$) therefore implies that ~140 km of right-slip has occurred on the Hosgri fault since the early Miocene. *Hall* [1975] has suggested that 80 to 100 km of right-lateral strike-slip movement has occurred on the Hosgri fault since the early Miocene, on the basis of similar stratigraphic sequences at Point Sal and San Simeon Point (Figure 1). However the 100-km offset between San Simeon Point and Point Sal does not represent the total slip experienced by the Hosgri fault, because the fault splays into the Santa Maria basin before reaching Point Sal [*Hall*, 1981*a*, *b*]. *Graham and Dickinson* [1978] have suggested that 115 km of displacement has occurred on the San Gregorio-Hosgri fault since the early Miocene, based primarily on the apparent offsets of geologic features along the northern segment of the fault. Assuming that their estimate of slip applies to the southern segment of the fault (that is, the fault does not bifurcate near Monterey Bay), the 140 km of slip predicated from a 90° rotation fo the Santa Ynez Range exceeds the observed slip by 25 km.

The additional 25 km of slip predicted by the above equation may have taken place on other NW trending faults adjacent to the Hosgri fault, such as the Naciemiento fault [*Page*, 1970] and offshore faults to the west of the Hosgri fault [*Hos-*

kins and Griffiths, 1971]. Alternatively, the additional displacement may have taken place by continuum shear within the Coast Ranges. Highly variable clockwise rotation of volcanic plugs and domes which intrude the Franciscan melange near San Luis Obispo [*Greenhaus and Cox*, 1979] suggests that some continuum shear has occurred within the Sur-Obispo belt between the Nacimiento and Hosgri faults (Figure 1).

Paleomagnetic data collected between the city of Santa Barbara and the San Gabriel fault (Figure 1) suggests that the eastern Santa Ynez Range has rotated approximately 60° clockwise since the early Miocene. (This eastward decrease in net tectonic rotation is due to post-Miocene oroclinal bending of the Santa Ynez Range between Santa Barbara and Ventura [see *Hornafius*, 1984*a*].) The predicted right-lateral strike-slip displacement within the Coast Ranges to the north of the eastern end of the Santa Ynez Range ($l = 70$ km) is 70 km, using equation 1. Neogene right-lateral displacements of 56 km on the Rinconada fault [*Dibblee*, 1976] and 14 km on the San Juan-Chimineas fault [*Bartow*, 1978] are consistent with a 60° clockwise rotation of the eastern Santa Ynez Range since the early Miocene. The total predicted strike-slip across the entire Coast Range province (assuming a 90° clockwise rotation of the western Santa Ynez Range and a 60° clockwise rotation of the eastern Santa Ynez Range) is 210 km, which is close to the suggested displacement of 185 km (Table 4). The agreement between the predicted and suggested right-lateral shear within the Coast Ranges since the early Miocene is a strong argument for the validity of the *Freund* [1974] and *Luyendyk et al.* [1980] model.

The timing of displacement on the Hosgri fault and adjacent Coast Range faults can be determined from the declination versus time curve for the Santa Ynez Range, since rotation and NW trending right-lateral fault displacement are related. The declination versus time curve in Figure 6 implies that the majority of displacement on these Coast Range faults occurred during the middle Miocene and Plio-Pleistocene, since these are the times of most rapid crustal rotation of the Santa Ynez Range. A 56° rotation of the Santa Ynez Range between 15 and 10 m.y. implies that ~85 km of right-slip occurred on the Hosgri fault system during the middle Miocene. An additional ~45 km of strike-slip in the Hosgri fault during the Plio-Pleistocene is implied by a 30° post-Miocene tectonic rotation of the Santa Ynez Range. Although poorly constrained, the amount of upper Miocene right-lateral strike-slip displacement on the Hosgri fault is predicted to be small (10 ± 40 km).

CONCLUSIONS

During the Neogene, the continental margin in southern California was subjected to dextral shear, due to the establishment of a transform boundary between the North American and Pacific plates [*Atwater*, 1970]. Within the California Coast Ranges, this dextral shear regime resulted in right-lateral strike-slip motion along parallel NW trending faults. However, at the latitude of the Transverse Ranges, the continental margin responded to the distributed dextral shear by coherent clockwise rotation of large crustal blocks. The southward termination of Neogene right-lateral strike-slip on the northwest-trending faults in the Coast Ranges was accomplished by splaying of the faults against the rotating northern boundary of the western Transverse Ranges province. Coherent clockwise rotation of the Santa Ynez Range during the Neogene thus enabled the northwest-trending faults in the Coast Ranges to experience ~200 km of post-early Miocene right-lateral strike-slip displacement, without cross-cutting the western Transverse Ranges province.

Paleomagnetic data collected from the Monterey Formation suggests that the Santa Ynez Range has rotated $95 \pm 9°$ clockwise since 15.2 ± 0.1 m.y. ago (Table 2). The rotation appears to have occurred in two stages (Figure 6). Between 15.2 ± 0.1 and 10 ± 1 m.y. ago, the range rotated $56 \pm 10°$ clockwise. Since the Miocene, the Santa Ynez Range rotated an additional $35 \pm 17°$ clockwise [*Hornafius*, 1984*a*]. Although poorly constrained, the amount of late Miocene rotation experienced by the Santa Ynez Range appears to be small ($4 \pm 17°$ clockwise). The timing of rotation of the Santa Ynez Range suggests that the majority of displacement on the Hosgri fault (and adjacent Coast Range faults) occurred during the middle Miocene and Plio-Pleistocene.

A 90° clockwise rotation of the Santa Ynez Range during the Neogene implies that the Paleogene strata within the range was deposited along the oceanward margin of an initially north-south trending forearc basin. The Franciscan melange flooring the Santa Maria basin was initially located to the west of this north-south trending forearc basin, where it formed the trench-slope break during the early Tertiary. Paleogene unconformities within the present-day northwestern Santa Ynez Range may have resulted from subduction-related tectonic instability within the adjacent accretionary wedge.

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